Computer Science & IT



Data Structure & Programming

Queue

Lecture No. 01



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Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Expression Infix to prefix

How to convert bracket expression

Evaluation of postfix prefix

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Linear Queue

Slide



Topic : Question

$-5 \Rightarrow 5-$

#Q.

Compute the post fix equivalent of the following expression $3 \times \log (x + 1) - a / 2$

$= 3 * \log x1 + - a2/$

$= 3 * x1 + \log - a2/$

$= 3x1 + \log * - a2/$

$3x1 + \log * a2/ -$

priority queue Implementation Heap



Topic : Question

C-Language

Recursion

A function f defined on stacks of integers satisfies the following properties. $f(\phi) = 0$ and f(push (S,i)) = max (f(s), 0) +I for all stacks S and integers i. If a stack S contains the integers 2, –3, 2, –1, 2 in order from bottom to top, what is f(S)?

(A) 6

(B) 4

(C) 3

(D) 2



Topic : Question

A function f defined on stacks of integers satisfies the following properties. $f(\phi) = 0$ and f(push (S,i)) = max (f(s), 0) +I for all stacks S and integers i. If a stack S contains the integers 2, –3, 2, –1, 2 in order from bottom to top, what is f(S)?

(A) 6

(B) 4

(C) 3

(D) 2

Queue



Queue



Empty Queue

front = rear = -1

Queue



rear

1

front

FIFO

Queue







1

2

Queue



1 2 3

Queue



Queue

rear = rear + 1



1 2 3 4 5

Queue



Queue



front



2

Queue







Topic : Queue

1. Queue is 2 ended data structure
2. front & rear
3. rear Index used for insertion (Enqueue)
4. front Index used for deletion (Dequeue)
5. Queue is a FIFO Linear Data structure

Slide



Topic : Queue

void init ()

Initialize Empty queue

void Enqueue(int x)

Insert x in rear end of queue

int isEmpty()

return 1 if queue is Empty else (Non Empty - 0)

int dequeue ()

Delete element from frontend of queue.

Slide



Topic : Queue

```
#define Max 100
int front, rear;
int a[Max];
```

```
int isEmpty() {
    if (front == -1)
        return 1;
    else
        return 0;
}
```

```
Void init() {
    front = rear = -1;
}
```

Slide



Topic : Queue

```
void Enqueue(int data){
    if(rear == Max-1){
        printf("Queue is full");
        return;
    }
    rear = rear+1;
    a[rear] = data;
    if(front == -1)
        front=0;
}
```

Linear queue

a[++rear]=data

Slide



Topic : Queue

```
int dequeue ( ) {
    int data;
    if (front == -1) {
        printf("Queue is empty");
        return -1;
    }
    data = a[front];
    if (front == rear)
        front = rear = -1;
    else
        front = front + 1;
} return data;
```

if(isEmpty())

limit.h

INT_MIN

if Last element is deleted

Slide



Queue

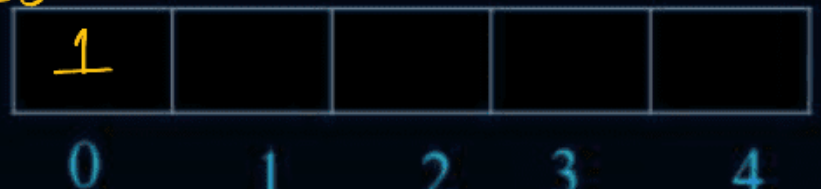
19:02

front = -1, rear = -1 Insert(1)



front = 0, rear = 0

First Insertion front will Increment to 0



front = 0, rear = 0



Insert(2)

fron



Slide

Queue





front = 0
rear = 1

Insert(3)

front = 0
rear = 2





front = 0
rear = 2

Insert(4)



front = 0
rear = 3

Slide

Queue



front = 0
rear = 3

Insert(4)



front = 0
rear = 4



front = 0
rear = 4

Insert(6)



front = 0
rear = 4

overflw

Slide

Queue



	2	3	4	5
0	1	2	3	4

front = 1

Insert(6)

overflow

	2	3	4	5
0	1	2	3	4

front = 1

rear = 4

No space available linearly

disadvantage: once rear index reaches to Max-1 the regardless of front queue is declared as full.

Slide



Queue

Dequeue

1	2	3	4	5
0	1	2	3	4

front = 0
rear = 4

	2	3	4	5
0	1	2	3	4

front = 1
rear = 4

	2	3	4	5
0	1	2	3	4

front = 1
rear = 4

dequeue

		3	4	5
0	1	2	3	4

front = 2
rear = 4

Slide

Queue

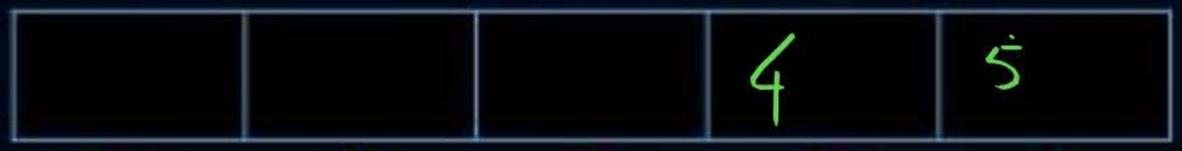


front = 2
rear = 4

dequeue()

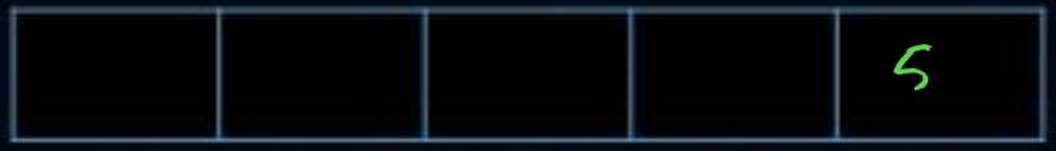


front = 3
rear = 4



front = 3
rear = 4

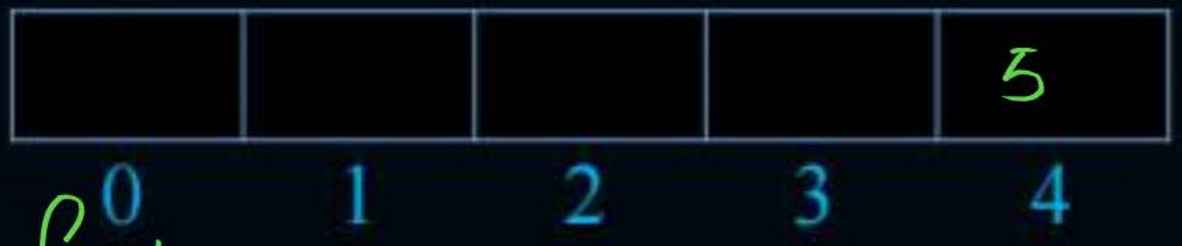
dequeue()



front = 4
rear = 4

Slide

Queue



front = 4
rear = 4

Dequeue



front = -1
rear = -1





after deletion before incrementing we check is front == rear

front = rear = -1

Slide

Topic: GATE 2021 Set-I, 1-Mark

Consider the following sequence of operations on an empty stack.

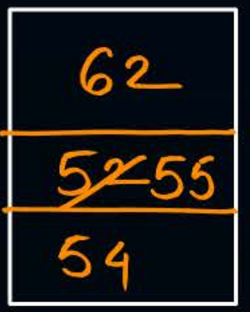
```
push (54); push (52); pop (); push (55); push (62); s = pop ();
```

Consider the following sequence of operations on an empty queue.

```
enqueue (21), enqueue (24), deque (); enqueue (28); enqueue (32); q = dequeue();
```

The value of s+q is 62+24 = 86

LIFO



S = pop()

S = 62

FIFO



q = dequeue()

q = 24

Slide

Question

If the following operation are performed on queue of size 5

I. Enqueue (11), Enqueue (12), Enqueue (13)

II. Dequeue(), Dequeue(), Enqueue(16), Enqueue(20), Enqueue (29), Enqueue (70), x= Dequeue ( ). If Y is Index of front & z is the Index of rear then X+Y-Z is

Enqueue(29)

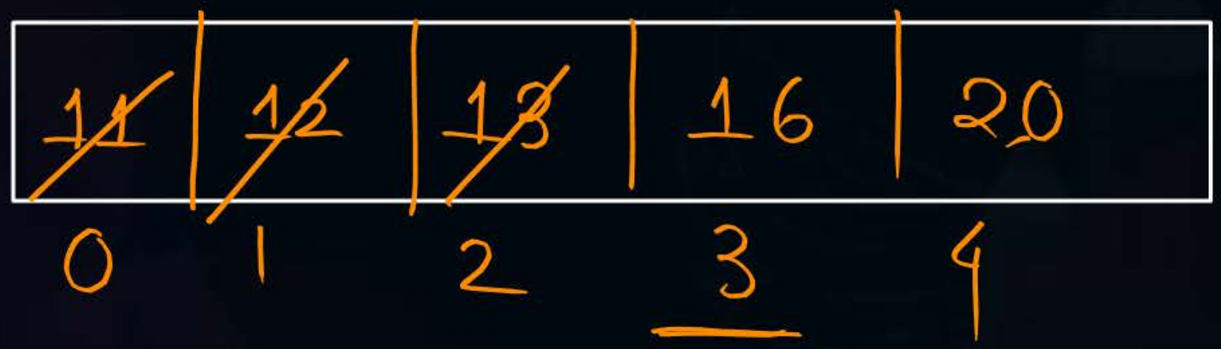
Enque(70)

↓

overflow

13+3−4

16−4 = 12



X = dequeue()

X = 13

Slide



Topic: GATE 2021 Set-I, 1-Mark

If the following operation are performed on queue of size 5

I. Enqueue (11), Enqueue (12), Enqueue (13)

II. Dequeue(), Dequeue(), Enqueue(16), Enqueue(20), Enqueue (29), Enqueue (70), x= Dequeue ( ). If Y is Index of front & z is the Index of rear then X+Y-Z is _____ 12 _____

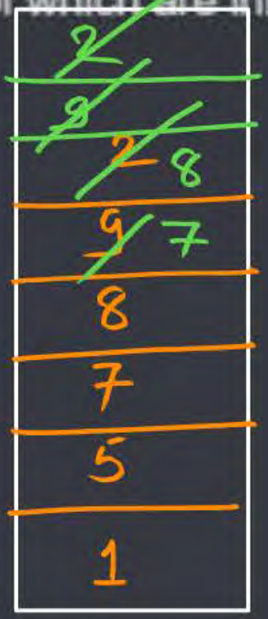
Slide

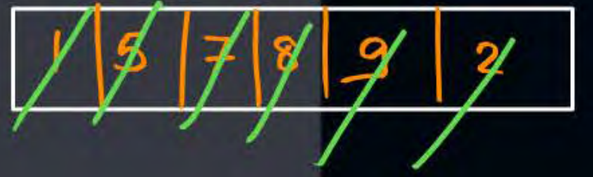
Topic: GATE 2023, 2-Marks

Consider a sequence a of elements $a_0 = 1, a_1 = 5, a_2 = 7, a_3 = 8, a_4 = 9$, and $a_5 = 2$. The following operations are performed on a stack S and a queue Q, both of which are initially empty.

I. push the elements of a from a_0 to a_5 in that order into S.
II. enqueue the elements of a from a_0 to a_5 in that order into Q.
III. pop an element from S.
IV. dequeue an element from Q.
V. pop an element from S.
VI. dequeue an element from Q.
VII. dequeue an element from Q and push the same element into S.
VIII. Repeat operation VII three times.
IX. pop an element from S.
X. pop an element from S.

1, 5, 7, 8, 9, 2





The top element of S after executing the above operations is _____.

[8]

Slide

SQ] [2-mark]

#Q. Let S1 and S2 be two stacks. S1 has capacity of 4 elements. S2 has capacity of 2 elements. S1 already has 4 elements: 100, 200, 300, and 400, whereas S2 is empty, as shown below.



Stack S1


Stack S2

(A) 100, 200, 400, 300 ×

(B) 200, 300, 400, 100

(C) 400, 200, 100, 300

(D) 300, 200, 400, 100

(B) pop to S_2
pop to S_2
GO 200
pop to S_1
GO 300
pop to S_1
GO 400
GO 100

Only the following three operations are available:
PushToS2: Pop the top element from S1 and push it on S2.
PushToS1: Pop the top element from S2 and push it on S1.
GenerateOutput: Pop the top element from S1 and output it to the user.
Note that the pop operation is not allowed on an empty stack and the push operation is not allowed on a full stack.
Which of the following output sequences can be generated by using the above operations?

SQ] [2-mark]

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Stack S1



Stack S2

(A) 100, 200, 400, 300 ✗

(B) 200, 300, 400, 100

(C) 400, 200, 100, 300

(D) 300, 200, 400, 100

GO 400
pop to S_2
GO - 200
GO 100
pop to S_1
GO

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PushToS2: Pop the top element from S1 and push it on S2.
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Stack S1


Stack S2

(A) 100, 200, 400, 300 ✗

(B) 200, 300, 400, 100 ✓

(C) 400, 200, 100, 300 ✓

(D) 300, 200, 400, 100 ✓

pop to S_2
GO - 300
GO - 200
pop to S_1
GO - 400
GO - 100

Only the following three operations are available:
PushToS2: Pop the top element from S1 and push it on S2.
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Which of the following output sequences can be generated by using the above operations?

A 100, 200, 400, 300

B 200, 300, 400, 100

C 400, 200, 100, 300

D 300, 200, 400, 100



2 mins Summary

Topic — Linear queue

Topic — Enqueue dequeue

Topic — problems

Topic

Topic

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t.me/Abhishekshaarmapw

THANK - YOU

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